

From a Deleted Bin to a Zero Arc:

Stable Padded Fourier Reassembly and the Low-Frequency Fixed-Shift Gate

Liang Wang

School of Artificial Intelligence and Automation

Huazhong University of Science and Technology, Wuhan 430074, P.R. China

wangliang.f@gmail.com

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Abstract

Let $A : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ have support S , let $D = \sum_n |A(n)|^2$, and let $Z = \widehat{A}(0) = \sum_n A(n)$. When $A(0) = 0$, Fourier inversion gives the exact identity

$$Z = - \sum_{r \neq 0} \widehat{A}(r).$$

Thus the distinguished zero mode can be recovered from the nonzero frequencies, but only as a coherent signed reassembly. We prove a stronger support-aware theorem. If $K = (\mathbb{Z}/q\mathbb{Z}) \setminus S$, then every weight w supported on K with $\sum_K w = 1$ gives

$$Z = - \sum_{r \neq 0} m_w(r) \widehat{A}(r), \quad m_w(r) = \sum_{k \in K} w(k) e^{2\pi i r k / q}.$$

The unique minimum- ℓ^2 kernel among zero-set averages is obtained by taking w uniform on K . Its squared norm is

$$\sum_{r \neq 0} |m_K(r)|^2 = \frac{|S|}{q - |S|},$$

and Parseval then recovers exactly the sharp physical inequality

$$|Z|^2 \leq |S|D.$$

Positive-density overpadding gives more. A fixed k -fold interval filter supported in the zero arc has Wiener norm $O_k(1)$ and multiplier decay

$$|m_k(r)| \ll_k d_q(r, 0)^{-k}.$$

For the literal TPC-32 packet this yields, unconditionally,

$$|Z_X| \ll_k \max_{0 < d_q(r, 0) \leq R} |\widehat{A}_X(r)| + X^{o(1)} N_{0,X} R^{1/2-k}.$$

At the critical row, the triangular case $k = 2$ reduces the full $q_X = X^{267/400+o(1)}$ frequency problem to the low window $R = X^{133/600+\varepsilon}$ for sufficiently small fixed $\varepsilon > 0$. We also derive an exact multiplier-weighted exceptional-frequency budget and prove that mean-square dispersion alone cannot improve the sharp support bound. Applied at one fixed nonzero shift, the result corrects an overly strong “nonzero frequencies cannot see the zero mode” formulation: coherent nonzero-frequency control can close the zero-mode gate, but generic frequency estimates and unweighted exceptional-set counts by themselves cannot. The Fourier theorems are L0, and their literal attachment is L1. No new L2 Möbius-correlation estimate is proved.

Keywords: distinguished zero mode; finite Fourier transform; coherent reassembly; exceptional frequencies; actual support; fixed shift; determinant coefficient.

Convention. Every arithmetic application uses one prescribed $h_0 \neq 0$, the literal high- β TPC-32 coefficient after complete determinant aggregation, its actual nonzero support, and its original global normalization. No assertion is specialized to $h_0 = 2$.

1 Introduction

The high-row fixed-shift route isolates a literal determinant coefficient

$$A_X(n) = A_{C_X}(n), \quad Z_X = \sum_n A_X(n), \quad D_X = \sum_n |A_X(n)|^2.$$

TPC-81 and TPC-82 record D_X and Z_X as the two remaining arithmetic statistics in the conditional flatness transfer [2, 3]. TPC-83 reduces natural determinant energy to survival of the actual post-bin ℓ^1 mass [4]. TPC-85 then audits several operations that do not manufacture cancellation: Fourier centering, determinant-fiber conditional expectation, and content inversion [5].

There is, however, a subtle correction to a common exclusion. The physical TPC-32 Fourier modulus is chosen large enough that the determinant support embeds without wrap, and the literal determinant bin $n = 0$ vanishes. Fourier inversion at that structural zero expresses Z_X as a sum of all nonzero Fourier modes. Consequently a sufficiently coherent theorem for those modes can control Z_X . What fails is the unproved passage from generic, mean-square, or all-but-exceptional frequency information to that particular signed sum.

This paper makes the distinction exact. Its main results are:

- (i) every zero coordinate gives an exact nonzero-frequency reconstruction of the zero mode;
- (ii) among kernels obtained by averaging structural zeros, the unique minimum- ℓ^2 kernel averages them uniformly;
- (iii) the optimal kernel and Parseval recover the sharp support inequality $|Z|^2 \leq |S|D$, with no loss;
- (iv) positive-density zero-arc padding produces fixed-order smooth filters with bounded Wiener norm and polynomial frequency decay;
- (v) the resulting tail estimate reduces the literal zero-mode problem to a mesoscopic low-frequency window;
- (vi) exceptional frequencies must be measured by their mass under the reconstruction multiplier, not merely by cardinality;
- (vii) total nonzero-frequency energy alone gives no improvement over the sharp support bound; and
- (viii) the literal TPC-32 packet satisfies all finite hypotheses, but none of the finite identities supplies new arithmetic cancellation.

The point is not to relabel the zero-mode problem as a Fourier problem. The exact identity instead says what a valid Fourier input must estimate. It also prevents two opposite errors: declaring that nonzero frequencies are logically irrelevant, and declaring that a generic dispersion estimate automatically controls their coherent sum.

1.1 Audit levels

We use the established three-level terminology. An L0 statement is finite algebra or an abstract countermodel. An L1 statement attaches that algebra to the literal fixed- h_0 carrier with complete native aggregation, actual support, and global normalization. An L2 statement is a new growing- X arithmetic estimate for that carrier. The main Fourier theorems below are L0; [Section 6](#) is L1; the missing coherent Möbius estimate is L2.

2 Exact reconstruction from structural zeros

Let $G = \mathbb{Z}/q\mathbb{Z}$, where $q \geq 2$, and write

$$e_q(x) = e^{2\pi i x/q}.$$

For $A : G \rightarrow \mathbb{C}$, use the unnormalized Fourier transform

$$\hat{A}(r) = \sum_{n \in G} A(n) e_q(-rn), \quad r \in G. \quad (1)$$

Then

$$A(n) = \frac{1}{q} \sum_{r \in G} \hat{A}(r) e_q(rn), \quad \sum_{r \in G} |\hat{A}(r)|^2 = q \sum_{n \in G} |A(n)|^2. \quad (2)$$

Put

$$Z = \hat{A}(0) = \sum_n A(n), \quad D = \sum_n |A(n)|^2.$$

Proposition 2.1 (One-zero coherent reconstruction). *If $A(k) = 0$ for some $k \in G$, then*

$$\boxed{Z = - \sum_{r \neq 0} \hat{A}(r) e_q(rk).} \quad (3)$$

In particular, if $A(0) = 0$, then

$$Z = - \sum_{r \neq 0} \hat{A}(r). \quad (4)$$

Proof. Insert $n = k$ in Fourier inversion and multiply by q :

$$0 = Z + \sum_{r \neq 0} \hat{A}(r) e_q(rk).$$

□

Remark 2.2 (What the identity does and does not say). [Proposition 2.1](#) proves that nonzero Fourier modes determine the zero mode once a structural zero is known. It does not bound the coherent sum on the right. A mean square, a pointwise estimate on generic r , or an exceptional-set count must still be combined with the exact phases $e_q(rk)$.

The single-zero identity is not optimal when A has many zero coordinates. Let

$$S = \text{supp } A, \quad s = |S|, \quad K = G \setminus S, \quad k = |K| = q - s.$$

Assume $K \neq \emptyset$. For a weight $w : K \rightarrow \mathbb{C}$ satisfying

$$\sum_{x \in K} w(x) = 1, \quad (5)$$

define

$$m_w(r) = \sum_{x \in K} w(x) e_q(rx). \quad (6)$$

Theorem 2.3 (Weighted zero-set reconstruction). *For every w satisfying (5),*

$$\boxed{Z = - \sum_{r \neq 0} m_w(r) \hat{A}(r).} \quad (7)$$

Moreover,

$$\sum_{r \neq 0} |m_w(r)|^2 = q \sum_{x \in K} |w(x)|^2 - 1. \quad (8)$$

Proof. Multiply the inversion identity $A(x) = 0$, $x \in K$, by $w(x)$ and sum over K . The $r = 0$ term is Z because $\sum_K w = 1$; the remaining terms give (7). Parseval applied to the zero-extension of w gives

$$\sum_{r \in G} |m_w(r)|^2 = q \sum_{x \in K} |w(x)|^2.$$

Since $m_w(0) = 1$, removing $r = 0$ proves (8). $\square$

3 The optimal support-averaged kernel

Theorem 3.1 (Unique minimum-norm reconstruction). *Among all complex weights w supported on K and satisfying $\sum_K w = 1$, the unique minimizer of*

$$\sum_{r \neq 0} |m_w(r)|^2$$

is the uniform weight

$$w_K(x) = \frac{1}{k}, \quad x \in K.$$

Writing $m_K = m_{w_K}$, one has

$$\boxed{\sum_{r \neq 0} |m_K(r)|^2 = \frac{s}{q - s}.} \quad (9)$$

Proof. Cauchy–Schwarz and $\sum_K w = 1$ give

$$1 \leq k \sum_{x \in K} |w(x)|^2,$$

with equality exactly when $w(x) = 1/k$ on K . Insert the minimum $\sum_K |w|^2 = 1/k$ into (8):

$$\min_w \sum_{r \neq 0} |m_w(r)|^2 = \frac{q}{k} - 1 = \frac{s}{q - s}.$$

$\square$

Theorem 3.2 (Optimal Fourier proof of the support bound). *For every $A : G \rightarrow \mathbb{C}$ with nonempty complement of its support,*

$$\boxed{|Z|^2 \leq sD.} \quad (10)$$

The constant s is sharp for every prescribed nonempty support set S .

Proof. Use [Theorem 2.3](#) with $w = w_K$ and apply Cauchy–Schwarz:

$$|Z|^2 \leq \frac{s}{q-s} \sum_{r \neq 0} |\hat{A}(r)|^2.$$

By Parseval,

$$\sum_{r \neq 0} |\hat{A}(r)|^2 = qD - |Z|^2.$$

Therefore

$$|Z|^2 \leq \frac{s}{q-s} (qD - |Z|^2),$$

which rearranges to (10). If $A(n) = c \neq 0$ on S and is zero off S , then $|Z|^2 = s^2|c|^2 = sD$, proving sharpness. $\square$

Remark 3.3 (No gain was hidden in Fourier space). The direct physical proof of (10) is Cauchy–Schwarz on S . The value of [Theorem 3.2](#) is diagnostic: it proves that the minimum-norm use of all structural zeros gives exactly, and not more than, the sharp physical support inequality. Any improvement must use arithmetic information about the coherent Fourier sum, not a different normalization of the same finite data.

Corollary 3.4 (Mean-square dispersion is insufficient). *Knowledge of s , D , and the exact nonzero-frequency energy*

$$\sum_{r \neq 0} |\hat{A}(r)|^2 = qD - |Z|^2$$

cannot yield a universal improvement over (10). Constant-modulus, common-phase coefficients on S attain equality.

Proof. The displayed identity is Parseval, and the extremizers in the proof of [Theorem 3.2](#) have the claimed data. $\square$

4 Stable reconstruction from a zero arc

The minimum- ℓ^2 kernel in [Theorem 3.1](#) is optimal for arbitrary zero sets. When the complement of the support contains a long cyclic interval, a different choice has bounded Wiener norm and polynomial Fourier decay.

Let $C \subset G$ be a cyclic interval of length H on which A vanishes. Fix an integer $k \geq 2$, independent of q , and choose L so that

$$k(L-1) + 1 \leq H. \tag{11}$$

Let u_L be the uniform probability measure on an interval of length L . Translate the k -fold cyclic convolution u_L^{*k} so that its support lies in C , and call the resulting probability measure p_k . Define

$$m_k(r) = \sum_{x \in C} p_k(x) e_q(rx).$$

Translation changes only the phase of m_k , while convolution gives

$$|m_k(r)| = \left| \frac{1}{L} \sum_{j=0}^{L-1} e_q(rj) \right|^k. \tag{12}$$

Theorem 4.1 (Stable zero-arc filter). *Assume $H \geq \delta q$ for a fixed $\delta > 0$, and choose L maximal subject to (11). For every fixed $k \geq 2$,*

$$Z = - \sum_{r \neq 0} m_k(r) \hat{A}(r), \quad (13)$$

$$\sum_{r \neq 0} |m_k(r)| \ll_{k,\delta} 1, \quad (14)$$

$$|m_k(r)| \ll_{k,\delta} (1 + d_q(r, 0))^{-k}, \quad (15)$$

where $d_q(r, 0) = \min(r, q - r)$ for the representative $0 \leq r < q$.

Proof. The first identity is Theorem 2.3 with $w = p_k$. Since $H \geq \delta q$ and k is fixed, maximality gives $L \asymp_{k,\delta} q$. The geometric-sum formula yields

$$\left| \frac{1}{L} \sum_{j=0}^{L-1} e_q(rj) \right| \leq \min \left\{ 1, \frac{q}{2L d_q(r, 0)} \right\},$$

after changing an absolute constant. Raising to the k -th power gives (15). Summing $\min(1, Cd^{-k})$ over the two frequencies at each distance d proves (14), because $k > 1$. $\square$

Proposition 4.2 (Exact triangular Wiener norm). *For $k = 2$,*

$$\sum_{r \neq 0} |m_2(r)| = \frac{q}{L} - 1. \quad (16)$$

Hence positive-density padding gives a constant Wiener norm.

Proof. By (12),

$$|m_2(r)| = |\hat{u}_L(r)|^2.$$

Parseval gives $\sum_r |\hat{u}_L(r)|^2 = q \|u_L\|_2^2 = q/L$. The $r = 0$ term is one. $\square$

Proposition 4.3 (Order-optimality of the Wiener norm). *Suppose coefficients c_r , $r \neq 0$, satisfy*

$$\hat{A}(0) = - \sum_{r \neq 0} c_r \hat{A}(r)$$

for every A supported in a fixed nonempty allowed arc. Then

$$\sum_{r \neq 0} |c_r| \geq 1.$$

Thus (14) has the optimal constant order.

Proof. Take A to be a point mass at any allowed coordinate n . Then $\hat{A}(0) = 1$ and $|\hat{A}(r)| = 1$, so the triangle inequality gives the claim. $\square$

Theorem 4.4 (Low-frequency reduction). *Under the hypotheses of Theorem 4.1, for $1 \leq R < q/2$,*

$$\boxed{|Z| \ll_{k,\delta} \max_{0 < d_q(r,0) \leq R} |\hat{A}(r)| + R^{1/2-k} \sqrt{qD}.} \quad (17)$$

Proof. On $0 < d_q(r, 0) \leq R$, use the maximum and (14). On the complementary frequencies, Cauchy–Schwarz, (15), and Parseval give

$$\begin{aligned} \sum_{d_q(r,0) > R} |m_k(r) \hat{A}(r)| &\leq \left(\sum_{d_q(r,0) > R} |m_k(r)|^2 \right)^{1/2} \left(\sum_r |\hat{A}(r)|^2 \right)^{1/2} \\ &\ll_{k,\delta} \left(\sum_{d > R} d^{-2k} \right)^{1/2} \sqrt{qD} \\ &\ll_{k,\delta} R^{1/2-k} \sqrt{qD}. \end{aligned}$$

□

Remark 4.5 (Why positive-density padding matters). If $H/q \rightarrow 0$, then $L/q \rightarrow 0$, and both the Wiener norm and the decay constant incur a factor depending on q/H . Such a loss must be entered into the physical ledger. The low-frequency theorem below therefore makes the harmless stronger auxiliary choice that leaves a zero arc of positive density.

5 A multiplier-weighted exceptional-frequency budget

Let $m = m_w$ be any exact reconstruction multiplier from Theorem 2.3. For $E \subseteq G \setminus \{0\}$, put

$$M_E^2 = \sum_{r \in E} |m(r)|^2, \quad H_E^2 = \sum_{r \in E} |\hat{A}(r)|^2,$$

and let E^c denote the complement inside $G \setminus \{0\}$.

Theorem 5.1 (Exact exceptional-frequency ledger). *For every $E \subseteq G \setminus \{0\}$,*

$$|Z| \leq M_E H_E + M_{E^c} H_{E^c}. \quad (18)$$

If in addition

$$|\hat{A}(r)| \leq U \quad (r \in E^c),$$

then

$$|Z| \leq M_E H_E + U \sum_{r \in E^c} |m(r)| \leq M_E H_E + U \sqrt{|E^c|} M_{E^c}. \quad (19)$$

Proof. Split the exact sum

$$Z = - \sum_{r \neq 0} m(r) \hat{A}(r)$$

over E and E^c . Apply Cauchy–Schwarz on both pieces to get (18). On E^c , the pointwise hypothesis gives the first bound in (19); a second Cauchy–Schwarz gives the last bound. □

Corollary 5.2 (A sufficient coherent transfer criterion). *Suppose a family A_X has target scale $T_X > 0$. If for some exceptional sets E_X ,*

$$M_{E_X} H_{E_X} = o(T_X) \quad \text{and} \quad M_{E_X^c} H_{E_X^c} = o(T_X),$$

then $Z_X = o(T_X)$. The same conclusion follows from the two terms on the right of (19).

Proof. Apply Theorem 5.1. □

Remark 5.3 (Cardinality is not the correct currency). The exceptional contribution is controlled by $M_E H_E$, not by $|E|$ alone. A small set can carry a large fraction of the reassembly multiplier or of the Fourier energy. Conversely a large set may be harmless if one of those weighted masses is small. Accordingly, an “all but few frequencies” theorem needs an explicit multiplier-weighted return to the literal coefficient.

Example 5.4 (A bounded exceptional set can carry the zero mode). Let q be a large prime, put $s = \lfloor q/8 \rfloor$, and define

$$A(n) = e_q(n) \mathbf{1}_{\{1, \dots, s\}}(n).$$

The coefficient has a zero arc of positive density and $A(0) = 0$. Its zero mode is a geometric sum with

$$|Z| \asymp s.$$

Moreover,

$$\widehat{A}(r) = \sum_{n=1}^s e_q((1-r)n),$$

so $\widehat{A}(1) = s$, while

$$|\widehat{A}(r)| \ll \min \left\{ s, \frac{q}{d_q(r, 1)} \right\}.$$

For every fixed $\varepsilon > 0$, only $O_\varepsilon(1)$ frequencies exceed εs , yet one of them carries the natural zero-mode scale. Therefore an all-but-bounded-exceptional-set pointwise theorem does not control Z unless the exceptional contribution is returned.

Proposition 5.5 (No automatic gain from the optimal zero-set kernel). *For the uniform zero-set multiplier m_K , using only*

$$M_E^2 + M_{E^c}^2 = \frac{s}{q-s}, \quad H_E^2 + H_{E^c}^2 = qD - |Z|^2,$$

the best universal consequence of (18) is the sharp support bound $|Z|^2 \leq sD$.

Proof. Cauchy–Schwarz in $\mathbb{C}^2$ gives

$$M_E H_E + M_{E^c} H_{E^c} \leq \left(\frac{s}{q-s} \right)^{1/2} (qD - |Z|^2)^{1/2}.$$

This is exactly the inequality used in [Theorem 3.2](#), and its rearrangement is $|Z|^2 \leq sD$. Equality is attained by the common-phase extremizers there, so no smaller universal constant follows. $\square$

6 The literal fixed-shift determinant interface

Fix one nonzero integer h_0 . On the high- β TPC-32 row,

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad C_X = \lfloor J_X \rfloor. \quad (20)$$

Let

$$A_X(n) = A_{C_X}(n)$$

be the literal matched normalized-determinant coefficient after all three physical channels and every native contribution in a determinant fiber have been summed. Write

$$S_X = \text{supp } A_X, \quad s_X = |S_X|, \quad Z_X = \sum_n A_X(n), \quad D_X = \sum_n |A_X(n)|^2.$$

TPC-32 proves

$$A_X(0) = 0, \quad s_X \ll Q_X, \quad (21)$$

and places the integer support in a fixed multiple of $[-Q_X, Q_X]$ [1].

Strengthen the harmless auxiliary padding choice. Put

$$Y_X = \lceil 4B_D Q_X + 4 \rceil$$

and use Bertrand's postulate to choose a prime

$$Y_X < q_X < 2Y_X. \quad (22)$$

The integer support is then injective modulo q_X , residue 0 represents only the literal bin $n = 0$, and $q_X \asymp Q_X$ may be arranged with constants depending only on the fixed determinant range. Since that containing arc has length at most $2B_D Q_X + 1 < q_X/2$, its complement has length $H_X > q_X/2$. This stronger choice changes no physical coefficient; it only adds zero coordinates to the auxiliary Fourier embedding.

Define the padded cyclic coefficient and its Fourier transform by

$$\tilde{A}_X(a) = \sum_{\substack{n \in \mathbb{Z} \\ n \equiv a \pmod{q_X}}} A_X(n), \quad \hat{\tilde{A}}_X(r) = \sum_{a \in \mathbb{Z}/q_X\mathbb{Z}} \tilde{A}_X(a) e_{q_X}(-ra). \quad (23)$$

No wrap gives

$$\sum_a \tilde{A}_X(a) = Z_X, \quad \sum_a |\tilde{A}_X(a)|^2 = D_X, \quad \tilde{A}_X(0) = A_X(0) = 0. \quad (24)$$

For readability, the remainder of this section writes $\hat{A}_X = \hat{\tilde{A}}_X$ and uses $\text{supp } A_X$ for the corresponding cyclic support.

Theorem 6.1 (Literal coherent-reassembly interface). *For the physical packet above,*

$$\boxed{Z_X = - \sum_{r=1}^{q_X-1} \hat{A}_X(r).} \quad (25)$$

If $K_X = (\mathbb{Z}/q_X\mathbb{Z}) \setminus \text{supp } A_X$ and

$$m_X(r) = \frac{1}{|K_X|} \sum_{k \in K_X} e_{q_X}(rk),$$

then

$$Z_X = - \sum_{r=1}^{q_X-1} m_X(r) \hat{A}_X(r), \quad (26)$$

$$\sum_{r=1}^{q_X-1} |m_X(r)|^2 = \frac{s_X}{q_X - s_X}, \quad (27)$$

$$|Z_X|^2 \leq s_X D_X. \quad (28)$$

The multiplier is determined by the actual zero set after complete physical aggregation.

Proof. The padding and (21) verify the hypotheses of Proposition 2.1 and Theorems 3.1 and 3.2. $\square$

Remark 6.2 (Correction to an earlier route exclusion). Equation (25) shows that a theorem for nonzero frequencies can control Z_X if it controls their exact coherent sum. Therefore “nonzero frequency” is not by itself a kill condition. What remains insufficient is a result that leaves an uncontrolled exceptional multiplier mass, averages away the fixed h_0 , replaces the literal phases or support, or supplies only unsigned mean-square dispersion.

6.1 Mesoscopic low-frequency reduction

The verified TPC-32 upper envelopes include

$$s_X \ll Q_X, \quad D_X \ll X^{o(1)} \frac{N_{0,X}^2}{Q_X}, \quad N_{0,X} = J_X Q_X^2.$$

Since $q_X \asymp Q_X$,

$$q_X D_X \ll X^{o(1)} N_{0,X}^2. \quad (29)$$

Apply the fixed-order zero-arc filter of [Theorem 4.4](#) to obtain, for $1 \leq R < q_X/2$,

$$|Z_X| \ll_k \max_{0 < d_{q_X}(r,0) \leq R} |\hat{A}_X(r)| + X^{o(1)} N_{0,X} R^{1/2-k}. \quad (30)$$

Theorem 6.3 (Critical-window criterion). *Write*

$$Q_X = X^{\beta+o(1)}, \quad J_X = X^{1-\beta+o(1)}$$

and fix $\eta_Z \geq 0$, $k \geq 2$, and $R_X = X^{\rho+o(1)} < q_X/2$. Suppose

$$\max_{0 < d_{q_X}(r,0) \leq R_X} |\hat{A}_X(r)| \leq X^{-\eta_Z+o(1)} Q_X^2 \quad (31)$$

and

$$(k - \frac{1}{2})\rho > 1 - \beta + \eta_Z. \quad (32)$$

Then

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2. \quad (33)$$

Proof. The first term of (30) has the required size by (31). The second, divided by the target scale, is

$$X^{o(1)} \frac{J_X Q_X^2 R_X^{1/2-k}}{X^{-\eta_Z} Q_X^2} = X^{1-\beta+\eta_Z-(k-1/2)\rho+o(1)},$$

which has a fixed-power saving under (32). $\square$

Corollary 6.4 (Triangular critical row). *At $\beta = 267/400$, with $\eta_Z = 0$ and $k = 2$, it is enough to prove (31) on*

$$R_X = X^{133/600+\varepsilon} \quad (34)$$

for every sufficiently small fixed $\varepsilon > 0$; explicitly one may take

$$0 < \varepsilon < \frac{267}{400} - \frac{133}{600} = \frac{107}{240}.$$

This window then satisfies $R_X < q_X/2$ for large X and is polynomially shorter than $q_X = X^{267/400+o(1)}$.

Proof. Here

$$(k - \frac{1}{2})\rho = \frac{3}{2} \left(\frac{133}{600} + \varepsilon \right) = \frac{133}{400} + \frac{3}{2}\varepsilon > 1 - \beta.$$

$\square$

Remark 6.5 (Higher fixed order). For every fixed $k \geq 2$, the sufficient window condition is

$$\rho > \frac{1 - \beta + \eta_Z}{k - 1/2}.$$

The order k must remain fixed unless every k -dependent constant is audited. Equality gives no fixed-power tail reserve.

6.2 What remains arithmetic

The support bound alone gives

$$|Z_X| \ll X^{o(1)} Q_X^2 J_X, \quad (35)$$

which is the natural absolute-mass scale. The TPC-32 zero-mode target is instead

$$|Z_X| \leq X^{o(1)} Q_X^2, \quad (36)$$

or a fixed-power improvement thereof. Thus finite reassembly does not create cancellation. Its advance is a rigorous localization: the missing arithmetic estimate can be confined to (31), while the high-frequency tail is absorbed by the already proved energy upper bound.

Corollary 6.6 (Exact form of a valid Fourier advance). *A nonzero-frequency argument closes (36) if it proves, for a literal reconstruction multiplier m ,*

$$\left| \sum_{r=1}^{q_X-1} m(r) \hat{A}_X(r) \right| \leq X^{o(1)} Q_X^2.$$

More generally, the low-window hypothesis of Theorem 6.3 or the two-term bound of Theorem 5.1 closes the target at the corresponding scale.

Proof. Use the exact reconstruction identity associated with m , then apply Theorems 5.1 and 6.3 as applicable. $\square$

Remark 6.7 (The minimal packet-level input). The pointwise maximum in (31) is a convenient sufficient condition, not the minimal one. The minimal new input bounds the complete packet-level signed sum

$$\sum_{0 < d_{q_X}(r,0) \leq R_X} m_k(r) \hat{A}_X(r)$$

at the target scale. Taking absolute values over native columns before their outer reassembly may lose a fixed power and must not be silently substituted.

Remark 6.8 (Affine twisted form and resonances). Expanding a literal low-frequency coefficient retains the TPC-85 fixed-shift affine pair

$$D_t = D_0 + st, \quad U_t = U_0 + at, \quad sU_0 - aD_0 = h_0,$$

now with the additive normalized-determinant character

$$e_{q_X} \left(-r \Delta_{\alpha, \gamma}^{\#}(j) \right).$$

Both polarizations, all three matched channels, the exact content cutoff, joint mask, and complete outer sum remain present. A nonzero r does not guarantee oscillation in every native column: resonant slopes, short fibers, and bounded fibers need their own return estimate. No such estimate is proved here.

Remark 6.9 (This is not an arithmetic estimate). Corollary 6.6 is an equivalence of interfaces. It does not prove the displayed upper bound. In particular it neither converts a generic dispersion theorem into a zero-mode estimate nor removes the fixed-shift affine Möbius correlation isolated in TPC-85.

7 Claim boundary and next gate

Theorem 7.1 (Audited conclusion). *For the actual high- β TPC-32 coefficient at one fixed nonzero h_0 , the distinguished zero mode is exactly a coherent linear functional of the nonzero determinant frequencies. Averaging over the complete actual zero set gives the unique minimum- ℓ^2 kernel within the zero-set-averaged class and recovers the sharp bound*

$$|Z_X|^2 \leq s_X D_X.$$

Positive-density zero-arc padding also gives (30); at the critical row the triangular filter reduces the missing pointwise input to $R_X = X^{133/600+\varepsilon}$ for sufficiently small fixed $\varepsilon > 0$. For any exceptional set, Theorem 5.1 gives the exact sufficient multiplier-weighted return condition.

Proof. Combine Theorems 4.4, 5.1, 6.1 and 6.3. □

What is proved. The finite reconstruction, optimizer, zero-arc filter, sharp support inequality, low-window reduction, and exceptional-frequency ledger are exact L0 results. Their application to the padded literal determinant coefficient and its inherited energy envelope is L1. The correction from “nonzero frequencies are irrelevant” to “a short low-frequency coherent reassembly is the remaining gate on the zero-mode branch” is therefore rigorous.

What is not proved. There is no new L2 estimate for the coherent sum, no lower bound for D_X , no fixed-shift Chowla theorem, and no transfer from averaged shifts or generic frequencies. The paper proves no parity breakthrough, no prime-pair lower bound, and no twin-prime result. Nothing is specialized to $h_0 = 2$.

Two independent ledgers. If a later theorem proves

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2,$$

then the TPC-82 flatness transfer uses

$$\lambda_D \leq 2\eta_Z.$$

Separately, every fixed-power degradation actually entering the full physical comparison must be counted once in

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

Equality in the second ledger stops the route; the first ledger is not another copy of the 1/400 budget.

Stop rules. The tail condition $(k - \frac{1}{2})\rho > 1 - \beta + \eta_Z$ is strict when a fixed-power reserve is claimed. If the padding has $H_X/q_X \rightarrow 0$, its filter loss must be recalculated. If k grows with X , its constants must be audited. An almost-all-frequency input is insufficient unless its multiplier-weighted exceptional contribution is returned. A resonant or short-fiber sector cannot be deleted.

Next gate. The positive route on the zero-mode branch is now precise: estimate the literal low-frequency multiplier sum, or its fixed- h_0 determinant-character-twisted affine Möbius expansion. A logarithmically averaged or almost-all-scale theorem may be used only after a quantitative Tauberian and quantifier transfer to the same growing packet. The next paper audits that transfer.

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